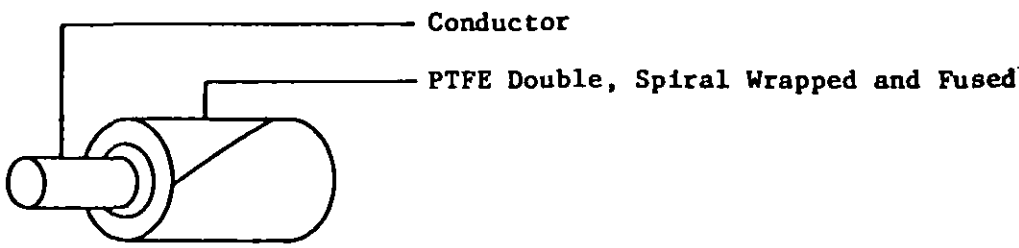
MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE (PTFE) INSULATED, 200°C, 1000 VOLTS, DOUBLE WRAPPED INSULATION

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this specification and the latest issue of MIL-W-16878.



SH 12090

FIGURE 1. Construction.

FSC 6145

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TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (max stranded) (inch)	Finished wire	
					Average diameter	
		Material	Coating		min (inch)	max (inch)
32	7 x 40	Copper	Nickel	0.010	0.036	0.044
30	7 x 38	Copper	Nickel	.012	.038	.046
28	7 x 36	Copper	Nickel	.015	.041	.049
26	7 x 34	Copper	Nickel	.019	.045	.053
24	7 x 32	Copper	Nickel	.024	.050	.058
22	7 x 30	Copper	Nickel	.030	.056	.064
20	7 x 28	Copper	Nickel	.038	.064	.072
18	7 x 26	Copper	Nickel	.048	.074	.084
16	19 x 29	Copper	Nickel	.057	.083	.095
14	19 x 27	Copper	Nickel	.072	.098	.114
12	19 x 25	Copper	Nickel	.091	.117	.133
10	37 x 26	Copper	Nickel	.111	.137	.153
8	133 x 29	Copper	Nickel	.169	.199	.219

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REQUIREMENTS:

Construction: The wire shall be insulated with two spirally wrapped layers of 0.005 inch (nominal) thick PTFE tape. The first layer of tape shall be applied in the opposite direction to the lay of the conductor and the second layer of tape applied in the opposite direction of the first layer, then heat bonded and fused.

Visual and mechanical inspection: Applicable.
Spark test: 5.0 kilovolts (kV).
Impulse dielectric test: 8.0 kV.
Tank test: 3.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$
Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20 degrees Celsius (°C).
K = 50,000
D = Maximum average diameter of finished wire
d = Conductor diameter
Conductor resistance: See table I.

Cold bend: Condition 4 hours at minus 65 ± 1°C

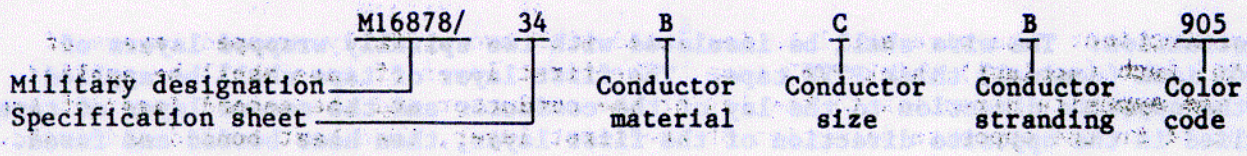
TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (max diameter) (inches)
32 through 16	1
14, 12	2
10, 8	3

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Applicable.
Heat resistance: Condition at 290°C.
Flammability: Not applicable.
Soldering: Applicable.
Heat aging: Not applicable.
Insulation tensile strength: 3000 pounds force per square inch (minimum).
Insulation elongation: 150 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

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Military part number (see MIL-W-16878):



Review activities:
Navy - AS, EC

User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N303)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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